Generative Modelling with Stochastic Differential Equations

Background: stochastic calculus and divergences

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A Background

In this appendix we collect the stochastic calculus and the divergence inequalities used in the course. For standard stochastic calculus we state the results and refer to [RY99, KS91, Øks03] for proofs. We prove the relative entropy bound between path laws (Proposition A.9) and the divergence inequalities of Section A.8.

Conventions. Vectors are columns. The norm $|\cdot|$ is Euclidean on vectors and Frobenius on matrices, so that $|A|^2 = \operatorname{tr}(AA^\top)$. Spatial derivatives act on x only. We write $C^{1,2}$ for functions of (t, x) that are continuously differentiable once in t and twice in x, and C_c^k for C^k functions of compact support. For a vector field $V = (V_1, \dots, V_n)$ in $C^1(\mathbb{R}^n; \mathbb{R}^n)$, the *divergence* is the function $\nabla \cdot V := \sum_{i=1}^n \partial_i V_i$, and $\Delta f = \nabla \cdot \nabla f = \sum_{i=1}^n \partial_i^2 f$ is the Laplacian.

A.1 Brownian motion and the Itô integral

Throughout, $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ is a filtered probability space satisfying the usual conditions. A process (X_t) is *adapted* if X_t is $\mathcal{F}_t$-measurable for every t. An adapted integrable process (M_t) is a *martingale* if $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for all $s \leq t$.

Definition A.1 (Brownian motion)**.** An *n-dimensional $(\mathcal{F}_t)$-Brownian motion* is an adapted process $(B_t)_{t\geq 0}$ in $\mathbb{R}^n$ such that $B_0 = 0$, the paths $t \mapsto B_t$ are continuous almost surely, and for all $0 \leq s < t$ the increment $B_t - B_s$ is independent of $\mathcal{F}_s$ and has law $\mathcal{N}(0, (t-s)\mathbb{I}_n)$.

The coordinates of B are independent one-dimensional Brownian motions. For $c > 0$ the process $(cB_{t/c^2})_{t\geq 0}$ is again a Brownian motion. An increment over a time Δt has size of order $\sqrt{\Delta t}$, so the squared increments do not sum to zero. More precisely, for partitions of $[0, t]$ with mesh tending to zero,

$$\sum_k \Delta_k B^i \, \Delta_k B^j \longrightarrow \delta_{ij}\, t \qquad \text{in } L^2(\mathbb{P}), \tag{A.1}$$

where $\Delta_k B = B_{t_{k+1}} - B_{t_k}$. This is summarised by the rules $\mathrm{d}B_t^i \, \mathrm{d}B_t^j = \delta_{ij}\, \mathrm{d}t$ and $\mathrm{d}t \, \mathrm{d}B_t^i = \mathrm{d}t \, \mathrm{d}t = 0$. These rules produce the second-order term in Itô's formula.

The Itô integral. For an adapted elementary process $H_t = \sum_k H_{t_k} \mathbf{1}_{(t_k, t_{k+1}]}(t)$ in $\mathbb{R}^{m\times n}$, we define $\int_0^t H_s \, \mathrm{d}B_s = \sum_k H_{t_k}(B_{t_{k+1}\wedge t} - B_{t_k \wedge t})$. Each coefficient is evaluated at a left endpoint, so it is known before the increment it multiplies. Approximation in $L^2(\mathrm{d}t \otimes \mathrm{d}\mathbb{P})$ extends the definition to progressively measurable H with $\mathbb{E}\int_0^T |H_s|^2 \, \mathrm{d}s < \infty$ [KS91, Section 3.2].

For such general H, left-endpoint Riemann sums need not converge to the integral. For instance, $H_t = \mathbf{1}_{\mathbb{Q}}(t)$ vanishes for almost every t, so its integral is zero. But its sums along any partition with rational left endpoints equal B_T. For continuous semimartingale integrands, left-endpoint sums converge in probability to the Itô integral. Midpoint sums give the Stratonovich integral instead.

Proposition A.2 (Itô isometry)**.** *Under these assumptions, $t \mapsto \int_0^t H_s \, \mathrm{d}B_s$ is a continuous square-integrable martingale with mean zero, and*

$$\mathbb{E}\left|\int_0^t H_s \, \mathrm{d}B_s\right|^2 = \mathbb{E}\int_0^t |H_s|^2 \, \mathrm{d}s, \qquad t \le T. \tag{A.2}$$

If only $\int_0^T |H_s|^2 \, \mathrm{d}s < \infty$ almost surely, the integral is still defined and is a continuous local *martingale, which need not have mean zero.*

For elementary integrands, (A.2) is a direct computation. The cross terms in the expanded square have conditional mean zero, since the later increment is independent of the rest of the product. The isometry is what makes the extension by approximation well defined.

A.2 Stochastic differential equations and Itô's formula

Let $b : [0,\infty) \times \mathbb{R}^n \to \mathbb{R}^n$ (the drift) and $g : [0,\infty) \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ (the diffusion coefficient) be measurable, and write $b_t(x) = b(t,x)$ and $g_t(x) = g(t,x)$. A *strong solution* of

$$\mathrm{d}X_t = b_t(X_t) \, \mathrm{d}t + g_t(X_t) \, \mathrm{d}B_t, \qquad X_0 = \xi, \tag{A.3}$$

with ξ $\mathcal{F}_0$-measurable (hence independent of the Brownian increments), is a continuous adapted process such that, almost surely, $X_t = \xi + \int_0^t b_s(X_s) \, \mathrm{d}s + \int_0^t g_s(X_s) \, \mathrm{d}B_s$ for all t, the integrals being well defined.

Theorem A.3 (Existence and uniqueness)**.** *Assume that for some $K < \infty$, all $t \in [0,T]$ and all $x, y \in \mathbb{R}^n$,*

$$|b_t(x) - b_t(y)| + |g_t(x) - g_t(y)| \le K|x-y|, \qquad |b_t(x)| + |g_t(x)| \le K(1+|x|).$$

Then (A.3) has a strong solution on $[0,T]$, unique up to indistinguishability. If $\mathbb{E}|\xi|^2 < \infty$, then $\mathbb{E}\sup_{t\le T}|X_t|^2 < \infty$.

See [Øks03, Theorem 5.2.1]. The drift of a reverse process can blow up at the final time T, and then we work on $[0, T-\delta]$ for some $\delta > 0$. Under the Lipschitz assumption of Theorem 5.1, the endpoint is also allowed.

Itô's formula is the chain rule for solutions of (A.3). Expanding $\varphi(t, X_t)$ by Taylor's formula, the products $\mathrm{d}X_t^i \, \mathrm{d}X_t^j$ contribute $(g_t g_t^\top)_{ij} \, \mathrm{d}t$ by (A.1). All other second-order terms are negligible.

Theorem A.4 (Itô's formula)**.** *Let X solve (A.3) and let $\varphi \in C^{1,2}$ be real-valued. Then, almost surely, for all $t \ge 0$,*

$$\mathrm{d}\varphi(t, X_t) = \Big(\partial_t \varphi + \langle b_t, \nabla\varphi\rangle + \tfrac{1}{2}\operatorname{tr}\big(g_t g_t^\top \nabla^2 \varphi\big)\Big)(t, X_t) \, \mathrm{d}t + \nabla\varphi(t, X_t)^\top g_t(X_t) \, \mathrm{d}B_t, \tag{A.4}$$

as an identity between integrals.

See [RY99, Theorem IV.3.3]. The formula also holds for general Itô processes $\mathrm{d}X_t = b_t \, \mathrm{d}t + \Sigma_t \, \mathrm{d}B_t$, with adapted b and Σ and with $\Sigma_t \Sigma_t^\top$ in place of $g_t g_t^\top$. It holds componentwise for vector-valued φ.

Further facts. The proof of Theorem 4.4 and some exercises use the following standard facts, which we state without proof [RY99, Chapters IV and V].

- For continuous local martingales M and N there is a unique continuous process $\langle M, N\rangle$ of finite variation, with $\langle M, N\rangle_0 = 0$, such that $MN - \langle M, N\rangle$ is a local martingale. It is the *quadratic covariation*, and $\langle M\rangle := \langle M, M\rangle$. For $M_t = \int_0^t H_r \,\mathrm{d}B_r$ with B a real Brownian motion, $\langle M\rangle_t = \int_0^t H_r^2 \,\mathrm{d}r$.
- *Itô's product rule.* For continuous semimartingales X and Y, $\mathrm{d}(X_tY_t) = X_t \,\mathrm{d}Y_t + Y_t \,\mathrm{d}X_t + \mathrm{d}\langle X, Y\rangle_t$, where $\langle X, Y\rangle$ is the quadratic covariation of their local martingale parts.
- A continuous local martingale of finite variation with $M_0 = 0$ is identically zero.
- *Lévy's characterisation.* A continuous local martingale M in $\mathbb{R}^n$ with $M_0 = 0$ and $\langle M^i, M^j\rangle_t = \delta_{ij}t$ is a Brownian motion with respect to the same filtration.
- *Deterministic time change.* If $t(\cdot)$ is a continuous increasing bijection of $[0, \infty)$ and M is a continuous local martingale, then $s \mapsto M_{t(s)}$ is a continuous local martingale in the filtration $(\mathcal{F}_{t(s)})_s$, with quadratic variation $\langle M\rangle_{t(s)}$.

A.3 Markov semigroups

A process (X_t) adapted to $(\mathcal{F}_t)$ is *Markov* if $\mathbb{E}[\varphi(X_t) \mid \mathcal{F}_s] = \mathbb{E}[\varphi(X_t) \mid X_s]$ for all $s \le t$ and all bounded measurable φ. In words, given the present, the past carries no further information about the future.

Assume that (A.3) is well posed, and write $(X_t^{s,x})_{t\ge s}$ for its solution started at time s from x. The *transition kernels* and the *Markov semigroup* are

$$p_{s,t}(x, A) = \mathbb{P}(X_t^{s,x} \in A), \qquad P_{s,t}\varphi(x) = \mathbb{E}\,\varphi(X_t^{s,x}) = \int \varphi(y)\, p_{s,t}(x, \mathrm{d}y), \qquad 0 \le s \le t.$$

The solution of (A.3) is a Markov process with $\mathbb{E}[\varphi(X_t) \mid \mathcal{F}_s] = P_{s,t}\varphi(X_s)$ [Øks03, Theorem 7.1.2]. The operators $P_{s,t}$ are linear and positive, satisfy $P_{s,s} = \mathrm{Id}$ and $P_{s,t}\mathbf{1} = \mathbf{1}$, and obey the *Chapman–Kolmogorov equations*

$$P_{r,t} = P_{r,s}P_{s,t}, \qquad r \le s \le t. \tag{A.5}$$

If b and g do not depend on time, then $P_{s,t} = P_{t-s}$ and $(P_t)_{t\ge0}$ is a one-parameter semigroup. On measures, $\nu P_{s,t}$ is defined by $\int \varphi \,\mathrm{d}(\nu P_{s,t}) = \int P_{s,t}\varphi \,\mathrm{d}\nu$, and

$$\mathrm{Law}(X_s) = \nu \quad\Longrightarrow\quad \mathrm{Law}(X_t) = \nu P_{s,t}. \tag{A.6}$$

Thus P acts on functions through expectations, and on measures through laws.

A.4 The generator

The *generator* of the semigroup is $\mathcal{L}_t\varphi = \lim_{h\downarrow0}(P_{t,t+h}\varphi - \varphi)/h$, defined for those φ for which the limit exists. For SDEs this includes $C_c^2(\mathbb{R}^n)$, which is all we need.

Proposition A.5 (Generator of an SDE)**.** *Let b and g be continuous and (A.3) well posed. For $\varphi \in C_c^2(\mathbb{R}^n)$,*

$$\mathcal{L}_t\varphi(x) = \langle b_t(x), \nabla\varphi(x)\rangle + \tfrac{1}{2}\operatorname{tr}\big(g_t(x)g_t(x)^\top \nabla^2\varphi(x)\big), \tag{A.7}$$

and, for $s \le t$ (Dynkin's formula),

$$\mathbb{E}\varphi(X_t) - \mathbb{E}\varphi(X_s) = \int_s^t \mathbb{E}\big[\mathcal{L}_r \varphi(X_r)\big]\,\mathrm{d}r. \tag{A.8}$$

More generally, if $\varphi \in C^{1,2}$ and the functions $\varphi(r,\cdot)$, $r \in [s,t]$, are supported in a fixed compact set, then

$$\mathbb{E}\varphi(t, X_t) - \mathbb{E}\varphi(s, X_s) = \int_s^t \mathbb{E}\big[(\partial_r + \mathcal{L}_r)\varphi(r, X_r)\big]\,\mathrm{d}r. \tag{A.9}$$

Sketch. By Itô's formula, $\varphi(X_t) - \varphi(X_s) - \int_s^t \mathcal{L}_r\varphi(X_r)\,\mathrm{d}r$ is a stochastic integral. Its integrand is bounded, since $\nabla\varphi$ has compact support and g is continuous. So it has mean zero, which is (A.8). Applying this to $X^{t,x}$ on $[t, t+h]$, dividing by h and letting $h \downarrow 0$ gives (A.7). □

With the generator, Itô's formula reads $\mathrm{d}\varphi(t, X_t) = (\partial_t + \mathcal{L}_t)\varphi(t, X_t)\,\mathrm{d}t + \mathrm{d}M_t$ with M a local martingale. Dynkin's formula with $X_s = x$ gives the *Kolmogorov forward equation* $\partial_t P_{s,t}\varphi = P_{s,t}\mathcal{L}_t\varphi$. When $P_{s,t}\varphi$ is C^2 in x, for instance for smooth bounded coefficients with bounded derivatives, one also has the *backward equation* $\partial_s P_{s,t}\varphi = -\mathcal{L}_s P_{s,t}\varphi$ [Øks03, Section 8.1]. In the time-homogeneous case, $\partial_t P_t\varphi = \mathcal{L}P_t\varphi = P_t\mathcal{L}\varphi$, so that formally $P_t = e^{t\mathcal{L}}$.

A.5 The Fokker–Planck equation

Dynkin's formula describes how expectations evolve. Its dual describes how the law of X_t evolves. The *adjoint* $\mathcal{L}_t^*$ acts on measures in the sense of distributions, by $\int \varphi\,\mathrm{d}(\mathcal{L}_t^*\nu) = \int \mathcal{L}_t\varphi\,\mathrm{d}\nu$ for $\varphi \in C_c^\infty(\mathbb{R}^n)$.

Proposition A.6 (Fokker–Planck equation)**.** *Let b and g be continuous, (A.3) well posed, and $\nu_t = \mathrm{Law}(X_t)$.*

(i) For every $\varphi \in C_c^2(\mathbb{R}^n)$, $t \mapsto \int \varphi\,\mathrm{d}\nu_t$ is C^1 and $\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi\,\mathrm{d}\nu_t = \int \mathcal{L}_t\varphi\,\mathrm{d}\nu_t$. That is, $\partial_t\nu_t = \mathcal{L}_t^*\nu_t$ in the sense of distributions.

(ii) If moreover $b_t \in C^1$ and $g_t \in C^2$ in x, and ν_t has a density ρ_t with $(t,x) \mapsto \rho_t(x)$ in $C^{1,2}$, then

$$\partial_t\rho_t = \mathcal{L}_t^*\rho_t = -\nabla\cdot(b_t\rho_t) + \frac{1}{2}\sum_{i,j=1}^n \partial_i\partial_j\big((g_t g_t^\top)_{ij}\rho_t\big). \tag{A.10}$$

Part (i) is Dynkin's formula, and part (ii) follows from (i) by integration by parts. For additive noise $g_t = \sqrt{2}\,\mathbb{I}_n$, (A.10) reads $\partial_t\rho_t = -\nabla\cdot(b_t\rho_t) + \Delta\rho_t = \nabla\cdot(\rho_t(\nabla\log\rho_t - b_t))$. The right-hand side vanishes at a density ρ when $b = \nabla\log\rho$.

Integration by parts on $\mathbb{R}^n$ produces no boundary term as soon as everything is integrable. The notes use this in the following form.

Lemma A.7 (No boundary term at infinity)**.** *Let $F \in C^1(\mathbb{R}^n;\mathbb{R}^n)$ with $\int|F|\,\mathrm{d}x < \infty$ and $\int|\nabla\cdot F|\,\mathrm{d}x < \infty$. Then $\int_{\mathbb{R}^n}\nabla\cdot F\,\mathrm{d}x = 0$.*

Proof. Fix $\chi \in C_c^\infty(\mathbb{R}^n)$ with $0 \le \chi \le 1$ and $\chi = 1$ on the unit ball, and set $\chi_R(x) := \chi(x/R)$, so that $\chi_R \to 1$ pointwise and $|\nabla\chi_R| \le \|\nabla\chi\|_\infty/R$. Since $\chi_R F$ has compact support, $\int \nabla\cdot(\chi_R F)\,\mathrm{d}x = 0$, and the product rule gives

$$\left|\int \chi_R\,\nabla\cdot F\,\mathrm{d}x\right| = \left|\int \langle F, \nabla\chi_R\rangle\,\mathrm{d}x\right| \le \frac{\|\nabla\chi\|_\infty}{R}\int|F|\,\mathrm{d}x \xrightarrow[R\to\infty]{} 0.$$

The left side tends to $|\int\nabla\cdot F\,\mathrm{d}x|$ by dominated convergence, with dominating function $|\nabla\cdot F|$. □

A.6 Invariant measures

Let (P_t) be time-homogeneous with generator $\mathcal{L}$. A probability measure μ is *invariant* if $\mu P_t = \mu$ for all $t \geq 0$, that is, if $X_0 \sim \mu$ implies $X_t \sim \mu$ for all t. If μ is invariant, then

$$\int \mathcal{L}\varphi \, \mathrm{d}\mu = 0 \quad \text{for all } \varphi \in C_c^\infty(\mathbb{R}^n), \qquad \text{that is,} \qquad \mathcal{L}^*\mu = 0, \tag{A.11}$$

since Dynkin's formula with $X_0 \sim \mu$ gives $0 = t \int \mathcal{L}\varphi \, \mathrm{d}\mu$. The converse holds under regularity assumptions. The case of the Langevin diffusion is Proposition 2.2.

A.7 Girsanov's theorem

Girsanov's theorem changes the drift of an SDE by reweighting the probability measure with an explicit density.

Theorem A.8 (Girsanov)**.** *Let B be an n-dimensional $(\mathcal{F}_t)$-Brownian motion under $\mathbb{P}$, and let $(\theta_t)_{t\in[0,T]}$ be a progressively measurable process in $\mathbb{R}^n$ satisfying* Novikov's condition

$$\mathbb{E}_{\mathbb{P}} \exp\Big(\frac{1}{2}\int_0^T |\theta_t|^2 \, \mathrm{d}t\Big) < \infty.$$

Then

$$\mathcal{E}_t = \exp\Big(\int_0^t \theta_s^\top \, \mathrm{d}B_s - \frac{1}{2}\int_0^t |\theta_s|^2 \, \mathrm{d}s\Big)$$

is a martingale on $[0,T]$ with $\mathbb{E}_{\mathbb{P}}\mathcal{E}_T = 1$, and under the probability measure $\mathbb{Q}$ with $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = \mathcal{E}_T$ the process $\tilde{B}_t = B_t - \int_0^t \theta_s \, \mathrm{d}s$ is an $(\mathcal{F}_t)$-Brownian motion.

The change of measure changes the drift and preserves the diffusion coefficient [KS91, Theorem 3.5.1 and Corollary 3.5.13]. For noise $\sqrt{2}\,\mathrm{d}B_t$, the choice $\theta_t = (b_t^1 - b_t^2)/\sqrt{2}$ changes drift b^2 to b^1. Since the density is explicit, so is the relative entropy between the two path laws. We compute it on $C_T = C([0,T];\mathbb{R}^n)$, with canonical process ω_t.

Proposition A.9 (Relative entropy between path laws)**.** *For $i = 1, 2$ let $b^i : [0,T] \times C_T \to \mathbb{R}^n$ be progressively measurable, so that $b_t^i(\omega)$ depends only on $(\omega_s)_{s\leq t}$, let ν^i be probability measures on $\mathbb{R}^n$, and let $\mathbb{Q}$ and $\mathbb{P}$ be the laws on C_T of solutions of*

$$\mathrm{d}X_t = b_t^1(X)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \ X_0 \sim \nu^1, \qquad \text{and} \qquad \mathrm{d}X_t = b_t^2(X)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \ X_0 \sim \nu^2.$$

Assume that the second SDE is well posed in law, that is, it has a solution, unique in law, for every initial law, also when started at a stopping time from a given initial segment of path. Assume moreover that $\mathbb{E}_{\mathbb{Q}} \int_0^T |b_t^1 - b_t^2|^2 \, \mathrm{d}t < \infty$. Then

$$\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}) \leq \mathrm{KL}(\nu^1 \,\|\, \nu^2) + \frac{1}{4}\,\mathbb{E}_{\mathbb{Q}} \int_0^T |b_t^1(\omega) - b_t^2(\omega)|^2 \, \mathrm{d}t, \tag{A.12}$$

with equality if, in addition, $\mathbb{E}_{\mathbb{Q}} \exp(\frac{1}{4}\int_0^T |b_t^1 - b_t^2|^2 \, \mathrm{d}t) < \infty$.

The proof applies Girsanov's theorem under $\mathbb{Q}$ to write the law of the second SDE as a reweighting of $\mathbb{Q}$. The relative entropy is then minus the expected logarithm of the density. Under the exponential moment condition, Novikov's condition holds and this works directly.

Proof under the exponential moment condition. Under $\mathbb{Q}$, $\mathrm{d}\omega_t = b_t^1\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t$ for a $\mathbb{Q}$-Brownian motion $\tilde{B}$. Apply Theorem A.8 under $\mathbb{Q}$ with $\theta_t = (b_t^2 - b_t^1)/\sqrt{2}$. Under $\mathrm{d}\mathbb{P}' = \mathcal{E}_T\,\mathrm{d}\mathbb{Q}$ the path solves the second SDE started from ν^1. By uniqueness in law, $\mathbb{P}'$ is the law of that SDE started from ν^1. Hence $\mathrm{KL}(\mathbb{Q}\|\mathbb{P}') = -\mathbb{E}_{\mathbb{Q}} \log \mathcal{E}_T = \frac{1}{4}\mathbb{E}_{\mathbb{Q}} \int_0^T |b_t^1 - b_t^2|^2\,\mathrm{d}t$, because the stochastic integral has mean zero. The laws $\mathbb{P}'$ and $\mathbb{P}$ have the same conditional law given ω_0, so the chain rule (Lemma A.11) adds $\mathrm{KL}(\nu^1\|\nu^2)$. □

The factor $\frac{1}{4}$ is $\frac{1}{2}$ divided by the noise variance 2. For a common invertible diffusion coefficient g_t, the integrand becomes $\frac{1}{2}|g_t^{-1}(b_t^1 - b_t^2)|^2$.

Finite energy need not imply Novikov's condition. We therefore stop the drift difference when its energy reaches a level k, apply Girsanov's theorem to the stopped problem, and let $k \to \infty$. The stopping times reach T almost surely under $\mathbb{Q}$, but not necessarily under $\mathbb{P}'$, so the limit needs care.

Proof of (A.12) under the finite-energy condition. Reductions. Let $\mathbb{P}'$ be the law of the second SDE started from ν^1. As in the previous proof, the chain rule gives $\mathrm{KL}(\mathbb{Q}\|\mathbb{P}) = \mathrm{KL}(\nu^1\|\nu^2) + \mathrm{KL}(\mathbb{Q}\|\mathbb{P}')$. So it suffices to show $\mathrm{KL}(\mathbb{Q}\|\mathbb{P}') \le C := \frac{1}{4}\mathbb{E}_{\mathbb{Q}} \int_0^T |\delta_t|^2\,\mathrm{d}t$, where $\delta_t = b_t^2 - b_t^1$. Write $(\mathcal{F}_t)$ for the canonical filtration on C_T, and let

$$\tau_k := \inf\Big\{t \le T : \int_0^t |\delta_s|^2\,\mathrm{d}s \ge k\Big\} \wedge T,$$

a stopping time, non-decreasing in k. Since $\int_0^T |\delta_s|^2\,\mathrm{d}s < \infty$ $\mathbb{Q}$-almost surely, the event $E := \{\tau_m = T \text{ for some } m\}$ has $\mathbb{Q}(E) = 1$.

Step 1, the hybrid laws. Under $\mathbb{Q}$, $\mathrm{d}\omega_t = b_t^1\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t$. Let $\theta_t^k = \delta_t \mathbf{1}_{t \le \tau_k}/\sqrt{2}$. Then $\int_0^T |\theta_t^k|^2\,\mathrm{d}t \le k/2$, so Novikov's condition holds. Theorem A.8 defines a probability $\mathbb{P}^k$ by $\mathrm{d}\mathbb{P}^k = \mathcal{E}_T^k\,\mathrm{d}\mathbb{Q}$, with $\mathcal{E}^k$ the stochastic exponential of θ^k. Under $\mathbb{P}^k$ the path solves $\mathrm{d}\omega_t = (b_t^1 + \delta_t \mathbf{1}_{t \le \tau_k})\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t^k$ for a Brownian motion $\tilde{B}^k$. So it follows the second SDE up to τ_k and the first afterwards. It starts from ν^1 because $\mathcal{E}_0^k = 1$. Since θ^k is square-integrable, $\int \theta^k\,\mathrm{d}\tilde{B}$ has mean zero under $\mathbb{Q}$, and

$$\mathrm{KL}(\mathbb{Q}\,\|\,\mathbb{P}^k) = -\mathbb{E}_{\mathbb{Q}} \log \mathcal{E}_T^k = \frac{1}{2}\mathbb{E}_{\mathbb{Q}} \int_0^T |\theta_t^k|^2\,\mathrm{d}t = \frac{1}{4}\mathbb{E}_{\mathbb{Q}} \int_0^{\tau_k} |\delta_t|^2\,\mathrm{d}t \le C.$$

Step 2, the hybrid laws agree with $\mathbb{P}'$ before τ_k. Under $\mathbb{P}^k$ the path solves the second SDE on $[0, \tau_k]$. Continue it after τ_k by a solution of the second SDE started from the path up to τ_k. The resulting law solves the second SDE on $[0, T]$ from ν^1, so it equals $\mathbb{P}'$ by uniqueness in law. It agrees with $\mathbb{P}^k$ on $\mathcal{F}_{\tau_k}$, since the continuation does not change the path before τ_k ([SV79, Chapter 6], [KS91, Section 5.4]). Hence $\mathbb{P}^k = \mathbb{P}'$ on $\mathcal{F}_{\tau_k}$.

Step 3, a uniform bound on $\mathcal{F}_{\tau_m}$. Fix m. For $k \ge m$, $\mathcal{F}_{\tau_m} \subseteq \mathcal{F}_{\tau_k}$, so $\mathbb{P}'$ and $\mathbb{P}^k$ agree on $\mathcal{F}_{\tau_m}$. Restricting to a smaller σ-algebra is data processing (Lemma A.12), so

$$\mathrm{KL}(\mathbb{Q}|_{\mathcal{F}_{\tau_m}} \,\|\, \mathbb{P}'|_{\mathcal{F}_{\tau_m}}) = \mathrm{KL}(\mathbb{Q}|_{\mathcal{F}_{\tau_m}} \,\|\, \mathbb{P}^k|_{\mathcal{F}_{\tau_m}}) \le \mathrm{KL}(\mathbb{Q}\,\|\,\mathbb{P}^k) \le C.$$

Step 4, passing to the limit. Let $\mathcal{G} = \sigma\big(\bigcup_m \mathcal{F}_{\tau_m}\big)$. Relative entropy along an increasing sequence of σ-algebras converges to the relative entropy on the σ-algebra they generate [PW25]. So $\mathrm{KL}(\mathbb{Q}|_{\mathcal{G}}\|\mathbb{P}'|_{\mathcal{G}}) \le C$, and in particular $\mathbb{Q} \ll \mathbb{P}'$ on $\mathcal{G}$, with a density Z. It remains to go from $\mathcal{G}$ to $\mathcal{F}_T$, and this is where the event E enters. For $A \in \mathcal{F}_T$ and every m, the set $A \cap \{\tau_m = T\}$ belongs to $\mathcal{F}_{\tau_m}$. So $A \cap E = \bigcup_m A \cap \{\tau_m = T\}$ belongs to $\mathcal{G}$. Since $\mathbb{Q}(E) = 1$,

$$\mathbb{Q}(A) = \mathbb{Q}(A \cap E) = \mathbb{E}_{\mathbb{P}'}\big[Z\,\mathbf{1}_{A \cap E}\big] \qquad \text{for every } A \in \mathcal{F}_T.$$

So on $\mathcal{F}_T$ we have $\mathbb{Q} \ll \mathbb{P}'$ with density $Z\mathbf{1}_E$, which is $\mathcal{G}$-measurable. Therefore $\mathrm{KL}(\mathbb{Q}\|\mathbb{P}') = \mathbb{E}_{\mathbb{Q}} \log(Z\mathbf{1}_E) = \mathbb{E}_{\mathbb{Q}} \log Z = \mathrm{KL}(\mathbb{Q}|_{\mathcal{G}}\|\mathbb{P}'|_{\mathcal{G}}) \le C$. □

The law $\mathbb{P}'$ may charge paths of infinite drift-difference energy, on which $\tau_k < T$ for every k. The event E removes them, and $\mathbb{Q}(E) = 1$. So no convergence of the stopping times under $\mathbb{P}'$ is needed; see also [CCL+23, Section 5.2].

A.8 Divergences between probability measures

Definition A.10 (Divergences)**.** Let μ, ν be probability measures on a Polish space, such as $\mathbb{R}^n$ or C_T.

- The *total variation distance* is $\mathrm{TV}(\mu,\nu) = \sup_A |\mu(A) - \nu(A)| = \frac{1}{2}\int |\frac{\mathrm{d}\mu}{\mathrm{d}\lambda} - \frac{\mathrm{d}\nu}{\mathrm{d}\lambda}|\,\mathrm{d}\lambda$, for any measure λ dominating both.
- The *relative entropy*, or Kullback–Leibler divergence, is $\mathrm{KL}(\mu\|\nu) = \int \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\,\mathrm{d}\mu$ if $\mu \ll \nu$, and $+\infty$ otherwise.
- The *χ^2-divergence* is $\chi^2(\mu\|\nu) = \int \big(\frac{\mathrm{d}\mu}{\mathrm{d}\nu} - 1\big)^2\,\mathrm{d}\nu$ if $\mu \ll \nu$, and $+\infty$ otherwise.
- The *Wasserstein distance* on a Polish space with metric d, between laws with finite second moments ($\int d(x,x_0)^2\,\mu(\mathrm{d}x) < \infty$ for some x_0), is $\mathcal{W}_2(\mu,\nu)^2 = \inf_\pi \int d(x,y)^2\,\pi(\mathrm{d}x,\mathrm{d}y)$, the infimum running over all couplings π of μ and ν. On $\mathbb{R}^n$, $d(x,y) = |x-y|$.
- The *relative Fisher information* on $\mathbb{R}^n$, for ν with a positive continuous Lebesgue density, is $\mathrm{FI}(\mu\|\nu) = \int |\nabla \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu}|^2\,\mathrm{d}\mu$ when $\mu \ll \nu$ and $\mathrm{d}\mu/\mathrm{d}\nu$ has a positive C^1 version, and $+\infty$ otherwise. Since ν charges every open set, $\mathrm{d}\mu/\mathrm{d}\nu$ has at most one continuous version, so the definition does not depend on a choice.

The assumption on ν in the last item is needed. For a singular ν, the gradient of $\mathrm{d}\mu/\mathrm{d}\nu$ would depend on the version chosen. With $\mu = \nu = \delta_0$, for instance, both 1 and $e^{\langle\theta,x\rangle}$ are positive smooth versions of $\mathrm{d}\mu/\mathrm{d}\nu$, and they give different values. For densities that vanish outside their support, Exercise 2.4 uses a different, support-based convention.

Relative entropy and χ^2 are of the form $D_\phi(\mu\|\nu) = \int \phi(\frac{\mathrm{d}\mu}{\mathrm{d}\nu})\,\mathrm{d}\nu$ for the convex functions $\phi(u) = u\log u$ and $\phi(u) = (u-1)^2$, with $\phi(1) = 0$. By Jensen's inequality, $D_\phi(\mu\|\nu) \geq \phi(\int \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\,\mathrm{d}\nu) = \phi(1) = 0$. When ϕ is strictly convex, equality holds if and only if $\mu = \nu$. Neither divergence is symmetric. TV and $\mathcal{W}_2$ are metrics.

Lemma A.11 (Chain rule)**.** *Let μ, ν be probability measures on a product $\mathcal{X} \times \mathcal{X}'$ of Polish spaces, with first marginals μ_1, ν_1 and conditional laws $\mu(\cdot \mid x), \nu(\cdot \mid x)$ of the second coordinate. Then*

$$\mathrm{KL}(\mu \,\|\, \nu) = \mathrm{KL}(\mu_1 \,\|\, \nu_1) + \int \mathrm{KL}(\mu(\cdot \mid x) \,\|\, \nu(\cdot \mid x))\,\mu_1(\mathrm{d}x).$$

In particular $\mathrm{KL}(\mu_1\|\nu_1) \leq \mathrm{KL}(\mu\|\nu)$.

Proof. Both sides are $+\infty$ unless $\mu_1 \ll \nu_1$ and $\mu(\cdot \mid x) \ll \nu(\cdot \mid x)$ for μ_1-almost every x. In that case $\mu \ll \nu$ with

$$\frac{\mathrm{d}\mu}{\mathrm{d}\nu}(x,y) = \frac{\mathrm{d}\mu_1}{\mathrm{d}\nu_1}(x)\,\frac{\mathrm{d}\mu(\cdot \mid x)}{\mathrm{d}\nu(\cdot \mid x)}(y).$$

Take the logarithm and integrate against $\mu(\mathrm{d}x,\mathrm{d}y) = \mu_1(\mathrm{d}x)\,\mu(\mathrm{d}y \mid x)$. See [PW25, Chapter 2] for the measure-theoretic details. □

For path laws one takes $x = \omega_0$, which is how the initial laws enter in Proposition A.9.

Lemma A.12 (Data processing)**.** *Let K be a Markov kernel and $D \in \{\mathrm{TV}, \mathrm{KL}, \chi^2\}$. Then $D(\mu K\|\nu K) \leq D(\mu\|\nu)$. In particular $D(F_\#\mu\|F_\#\nu) \leq D(\mu\|\nu)$ for every measurable map F, and $D(\mu P_{s,t}\|\nu P_{s,t}) \leq D(\mu\|\nu)$ for every Markov semigroup.*

Proof. For TV, $|\mu K(A) - \nu K(A)| = |\int K(x, A)\,(\mu - \nu)(\mathrm{d}x)| \le \mathrm{TV}(\mu, \nu)$, since $0 \le K(\cdot, A) \le 1$ and $\mu - \nu$ has total mass zero. For $D = D_\phi$, we may assume $\mu \ll \nu$ with $f = \mathrm{d}\mu/\mathrm{d}\nu$. Let (X, Y) have law $\nu(\mathrm{d}x)K(x, \mathrm{d}y)$. For bounded h, $\int h\,\mathrm{d}(\mu K) = \mathbb{E}[f(X)h(Y)] = \mathbb{E}\big[\mathbb{E}[f(X) \mid Y]\,h(Y)\big]$. So $\mu K \ll \nu K$ with density $\mathbb{E}[f(X) \mid Y]$. By conditional Jensen,

$$D_\phi(\mu K \| \nu K) = \mathbb{E}\,\phi\big(\mathbb{E}[f(X) \mid Y]\big) \le \mathbb{E}\,\phi(f(X)) = D_\phi(\mu \| \nu).$$

For a map F, take $K(x, \cdot) = \delta_{F(x)}$. □

In words, passing two laws through the same channel cannot make them easier to distinguish.

The bounds below on TV and on KL follow from Cauchy–Schwarz and Jensen. Pinsker's inequality reduces, by data processing, to an inequality between two Bernoulli laws.

Lemma A.13 (Comparison of divergences)**.** *For probability measures μ, ν,*

$$\mathrm{TV}(\mu, \nu) \le \sqrt{\tfrac{1}{2}\,\mathrm{KL}(\mu \| \nu)} \quad \textit{(Pinsker)}, \qquad \mathrm{TV}(\mu, \nu) \le \tfrac{1}{2}\sqrt{\chi^2(\mu \| \nu)},$$
$$\mathrm{KL}(\mu \| \nu) \le \log\big(1 + \chi^2(\mu \| \nu)\big) \le \chi^2(\mu \| \nu).$$

Proof. We may assume $\mu \ll \nu$, with $f = \mathrm{d}\mu/\mathrm{d}\nu$. By Cauchy–Schwarz, $\mathrm{TV}(\mu, \nu) = \frac{1}{2}\int |f - 1|\,\mathrm{d}\nu \le \frac{1}{2}\big(\int (f-1)^2\,\mathrm{d}\nu\big)^{1/2}$. By Jensen's inequality for the concave logarithm, $\mathrm{KL}(\mu \| \nu) = \int \log f\,\mathrm{d}\mu \le \log \int f\,\mathrm{d}\mu = \log \int f^2\,\mathrm{d}\nu = \log(1 + \chi^2(\mu \| \nu))$, and $\log(1 + u) \le u$.

For Pinsker's inequality, let $A = \{f > 1\}$, so that $\mathrm{TV}(\mu, \nu) = \mu(A) - \nu(A) = p - q$ with $p = \mu(A)$ and $q = \nu(A)$. By data processing for the map $\mathbf{1}_A$, $\mathrm{KL}(\mu \| \nu) \ge k(p, q) := p \log \frac{p}{q} + (1 - p) \log \frac{1-p}{1-q}$, the relative entropy between Bernoulli laws. If $q \in \{0, 1\}$ then $p = q$, since $\mu \ll \nu$, and there is nothing to prove. Otherwise, for fixed p, the function $g(q) = k(p, q) - 2(p - q)^2$ vanishes at $q = p$, and

$$g'(q) = -\frac{p}{q} + \frac{1-p}{1-q} + 4(p - q) = (q - p)\Big(\frac{1}{q(1-q)} - 4\Big),$$

which has the sign of $q - p$ because $q(1 - q) \le \frac{1}{4}$. So g is minimised at $q = p$, and $g \ge 0$. That is, $\mathrm{KL}(\mu \| \nu) \ge 2(p - q)^2 = 2\ \mathrm{TV}(\mu, \nu)^2$. □

Pinsker's inequality converts the relative entropy bounds of Chapter 5 into total variation bounds. The χ^2 bounds are useful when a Poincaré inequality controls χ^2, as in Chapter 3. By the last inequality, a bound in χ^2 is stronger than the same bound in KL. For product measures, relative entropy adds up while $1 + \chi^2$ multiplies. So the χ^2-divergence can be exponentially larger in high dimension.

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